

CSE422 Assignment 03

Answer to the Question No 01

The system mentioned in the case study is fitness tracking device and mobile application such as Fitbit. These applications and services collect and store huge amount of personal data like name, age, height, weight, location and date of birth of the user. These are used to monitor health, improve fitness and provide appropriate health advice. However, these can create some risks for society and environment.

The case of leaking 61 million user's data in 2021 shows us that how vulnerable the health data can be if the company fails to secure their database properly. Personal information like name, age, date of birth, height, weight and geographic location can be used as fake identity to perform cybercrime. As the health data is very personal therefore exposure of them can cause mental and social trauma to the exposed user. This can reduce their trust in digital health technologies.

In long term, the dominating tech company's continuous data collection takes them toward surveillance capitalism. The user's personal information and behavior then used as economic asset. Zuboff (2019) explains that these give the companies to predict and influence the behavior of the user. Google's acquisition of Fitbit rises concern of Fitbit's health data being combined with Google's advertise system. This could strengthen the monopoly market and decrease competition (Bourreau et al., 2020).

Moreover, this system is unstable in terms of environmental side. Huge amount of data storage and analysis needs larger data centers. They use a large amount of electricity and increase carbon emission. At the same time, to run AI related ads and analysis system more energy is required. That's why, without proper control these technologies are not environmentally eligible for long run.

Crawford (2021) mentions that AI systems use a large amount of data and computational resources ignoring environmental and human cost. Therefore, though the health tracking technology is beneficial but without proper monitoring and control they are socially and environmentally risky.

Answer to the Question No 02

This system raises ethical issues regarding privacy, transparency and accountability. Although in the case study no algorithmic bias was mentioned but the use of health data can lead to discrimination in ads or decision making.

One major problem is the lack of transparency. Most of the users do not know how their data are being collected, used and sold. Usually, the privacy policies of the companies are complex and long. For this the user often lets their data be used without even reading or understanding the policy properly (European Commission, 2020).

Another big problem is accountability. The leak of millions of data shows how the companies did not take proper security measures. Though the data was later secured but it tells that the security measures were weak. They should have proper security system long before the data leak.

There is also the risk of biasness and discrimination. Obermeyer et al. (2019) have shown that the health related algorithm sometime create discrimination against socially disadvantaged people because of biased data. If wearable device is combined with Ai based profiling system then some uses may face unfair treatment in insurance, employment and healthcare.

Zuboff (2019) also argues that surveillance capitalism reduces the personal freedom of people. Because large companies gain excessive control over people's behavior. Which create imbalance of power between the user and the company.

Answer to the Question No 03

If this health related information based system is used in real world it would face various legal issues like privacy and data protection laws. Health information is very sensitive personal information. So, the company must abide by strict laws.

According to the General Data Protection Regulation (GDPR) of European Union, without the proper consent of the user personal information can not be collected and there must be strong security measures for data protection (GDPR, n.d.). Also, California Consumer Privacy Act (CCPA) gives the user the right to control and delete their personal information (CCPA, n.d.).

Google's Fitbit acquisition was legally controversial because it could create risks for competition and user privacy. Although the acquisition was approved Google had to comply with certain conditions (European Commission, 2020).

Engineers have the responsibility to design the system in such way that they are safe, fair and lawful. This requires strong cyber security, privacy and bias free Ai. Floridi et al. (2018) mentioned that justice, autonomy and transparency is must requirement in ethical Ai system.

Violating the law and ethical guidelines can cause fines, lawsuits and loss of trust to the company. Therefore, engineers play very important role in developing responsible Ai.

References:

- Bourreau, M., de Streel, A., & Graef, I. (2020). *Big Tech acquisitions: Competition & innovation effects and EU merger control*. CERRE. https://cerre.eu/wp-content/uploads/2020/03/cerre_big_tech_acquisitions_2020.pdf
- Crawford, K. (2021). *Atlas of AI: Power, politics, and the planetary costs of artificial intelligence*. Yale University Press. https://books.google.com/books?hl=en&lr=&id=zdu0EAAAQBAJ&oi=fnd&pg=PA110&dq=Atlas+of+AI+Kate+Crawford&ots=Nk7kFubqRy&sig=PiqZuPju2PssImXML9DlIB_zLUg
- European Commission. (2020). *Commission clears acquisition of Fitbit by Google, subject to conditions*. https://ec.europa.eu/commission/presscorner/detail/en/ip_20_2484
- Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., ... & Vayena, E. (2018). AI4People-An ethical framework for a good AI society: Opportunities, risks, principles, and recommendations. *Minds and Machines*. <https://link.springer.com/article/10.1007/S11023-018-9482-5>
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*. <https://www.science.org/doi/abs/10.1126/science.aax2342>
- Zuboff, S. (2019). *The age of surveillance capitalism: The fight for a human future at the new frontier of power*. PublicAffairs. <https://api.taylorfrancis.com/content/chapters/edit/download?identifierName=doi&identifierValue=10.4324/9781003320609-27&type=chapterpdf>